WORKED SOLUTION

STEP I 2008, Question 1

Pure • Proof by Contradiction

> **STEP I 2008, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/08-s1-q1

The key idea

Rational numbers are closed under addition, subtraction, multiplication and division by a non-zero number. Almost everything here follows from that one fact: if some numbers are rational, anything you build from them with these operations is rational too. In the final part, assume two of the four numbers are rational, combine them cleverly, and produce one of the numbers we were told is irrational.

Solution

Definition

A real number x is *rational* if it can be written as $x = \frac{m}{n}$ with m and n integers and $n \neq 0$. A real number is *irrational* if it is not rational, that is, if there are no integers m and $n \neq 0$ with $x = \frac{m}{n}$.

Statement A

Suppose, for a contradiction, that pq is irrational but neither p nor q is irrational. Then both are rational, so we can write $p = \frac{a}{b}$ and $q = \frac{c}{d}$ with a, b, c, d integers and $b, d \neq 0$. Then

$$pq = \frac{ac}{bd},$$

where ac and bd are integers and $bd \neq 0$ because neither b nor d is zero. So pq is rational, which contradicts our assumption. Hence at least one of p and q is irrational.

Statement B

In the same way, suppose $p + q$ is irrational but $p = \frac{a}{b}$ and $q = \frac{c}{d}$ are both rational, with $b, d \neq 0$. Then

$$p + q = \frac{ad + bc}{bd},$$

a ratio of integers with non-zero denominator, so $p+q$ is rational. This contradiction shows that at least one of p and q is irrational.

Statement C

Take $p=\sqrt{2}$ and $q=-\sqrt{2}$. Both are irrational, but $p+q=0=\frac{0}{1}$ is rational, so C is false.

For completeness, here is why $\sqrt{2}$ is irrational. If $\sqrt{2}=\frac{m}{n}$ in lowest terms, then $m^2=2n^2$, so m^2 is even and therefore m is even (the square of an odd number is odd). Writing $m=2k$ gives $n^2=2k^2$, so n is even too, which contradicts lowest terms. And $-\sqrt{2}$ is irrational because if $-\sqrt{2}=\frac{m}{n}$ then $\sqrt{2}=\frac{-m}{n}$.

The final part

We are told that π, π^2 and $\mathrm{e}\pi$ are irrational (these three facts are all we need). Two simple observations will be used repeatedly:

- if r and s are rational, then so are $r+s$, $r-s$, rs, and r/s when $s\neq 0$ (the argument is exactly the one used for A and B);
- consequently, if r is rational, then so is $\frac{1}{2}r$, since $\frac{1}{2}$ is rational.

Note also that $\pi+\mathrm{e}>0$ and $\pi-\mathrm{e}>0$, since $\pi>3>\mathrm{e}>0$; in particular neither is zero.

Suppose, for a contradiction, that two of the four numbers

$$\pi+\mathrm{e},\quad \pi-\mathrm{e},\quad \pi^2-\mathrm{e}^2,\quad \pi^2+\mathrm{e}^2$$

are rational. There are six possible pairs, and we rule out each one.

1. $\pi+\mathrm{e}$ *and* $\pi-\mathrm{e}$ *rational.* Adding, 2π is rational, so π is rational. Contradiction.
2. $\pi^2-\mathrm{e}^2$ *and* $\pi^2+\mathrm{e}^2$ *rational.* Adding, $2\pi^2$ is rational, so π^2 is rational. Contradiction.
3. $\pi+\mathrm{e}$ *and* $\pi^2+\mathrm{e}^2$ *rational.* Squaring the first, $(\pi+\mathrm{e})^2$ is rational, and
$$2\mathrm{e}\pi=(\pi+\mathrm{e})^2-(\pi^2+\mathrm{e}^2),$$
so $\mathrm{e}\pi$ is rational. Contradiction.
4. $\pi-\mathrm{e}$ *and* $\pi^2+\mathrm{e}^2$ *rational.* Similarly,
$$2\mathrm{e}\pi=(\pi^2+\mathrm{e}^2)-(\pi-\mathrm{e})^2$$
is rational, so $\mathrm{e}\pi$ is rational. Contradiction.
5. $\pi+\mathrm{e}$ *and* $\pi^2-\mathrm{e}^2$ *rational.* Since $\pi+\mathrm{e}\neq 0$,
$$\pi-\mathrm{e}=\frac{\pi^2-\mathrm{e}^2}{\pi+\mathrm{e}}$$

is a quotient of rationals with non-zero denominator, so it is rational. Now $\pi + \mathrm{e}$ and $\pi - \mathrm{e}$ are both rational, which is impossible by case 1.

6. $\pi - \mathrm{e}$ *and* $\pi^2 - \mathrm{e}^2$ *rational.* Since $\pi - \mathrm{e} \neq 0$,

$$\pi + \mathrm{e} = \frac{\pi^2 - \mathrm{e}^2}{\pi - \mathrm{e}}$$

is rational, and again case 1 gives a contradiction.

Ask yourself Why do we need $\pi - \mathrm{e} \neq 0$ in case 6?

Because dividing by zero is not allowed: the step "rational divided by rational is rational" only works when the denominator is non-zero. If we had $\pi = \mathrm{e}$, then $\pi^2 - \mathrm{e}^2 = 0$ would be rational without telling us anything about $\pi + \mathrm{e}$. Here $\pi - \mathrm{e} \approx 0.42$, so all is well.

Every pair leads to a contradiction, so no two of the four numbers can both be rational. Hence at most one of them is rational, as required.

Checks

The identities used are easy to confirm numerically: with $\pi \approx 3.14159$ and $\mathrm{e} \approx 2.71828$, we get $(\pi + \mathrm{e})(\pi - \mathrm{e}) \approx 5.85987 \times 0.42331 \approx 2.48055 \approx \pi^2 - \mathrm{e}^2$, and $(\pi + \mathrm{e})^2 - (\pi^2 + \mathrm{e}^2) \approx 17.0795 \approx 2\mathrm{e}\pi$. Count the pairs as well: four numbers give $\binom{4}{2} = 6$ pairs, and all six are covered. Notice that the irrationality of e and e^2 was never used; by symmetry we could have used them in place of π and π^2 in cases 1 and 2.

Common slips

- Proving the converse of A or B by mistake. A says "product irrational $\Rightarrow$ a factor irrational"; its converse is false ($\sqrt{2} \cdot \sqrt{2} = 2$).
- Dividing by $\pi \pm \mathrm{e}$ without remarking that it is non-zero.
- Using statement C as if it were true, for example claiming "π and e irrational, so $\pi + \mathrm{e}$ is irrational". Whether $\pi + \mathrm{e}$ is irrational is in fact an open problem.

Beyond STEP

The closure facts used here say that the rationals form a *field*: a set where you can add, subtract, multiply and divide (except by zero) without leaving it. Fields are a central object in a first-year algebra course, and questions such as "is $\pi + \mathrm{e}$ irrational?" lead into transcendence theory, where remarkably little is known. For the algebra, try Allenby, *Rings, Fields and Groups*; for the number theory of rationals and irrationals, Davenport, *The Higher Arithmetic*.

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